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Collaborative Fact-checking via Multi-Agent Debate and Weighted Voting

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***Abstract*—This paper introduces a novel multi-agent collaborative framework for fact-checking, leveraging debate and weighted voting to enhance accuracy and reliability. Addressing the limitations of single large language model (LLM) approaches and sequential multi-agent methods, our framework facilitates dynamic knowledge sharing and collaboration among heterogeneous agents. By simulating real-world argumentation through group debates and employing a dynamically adjusted weighted voting mechanism, we effectively integrate diverse agent expertise. Our experimental results on publicly available benchmarks show improvements over baseline models, with absolute increases of 21.9 percentage points in accuracy, 11.2 percentage points in precision, and 22.6 percentage points in recall. This research highlights the efficacy of multi-agent collaboration in improving fact-checking performance, offering a robust solution to combat the spread of misinformation in the digital age.**

Keywords-fact-checking, multi-agents, LLM, collaborative

I. Introduction

The proliferation of misinformation, fake news, and misleading content poses a major challenge in the digital age. The importance of fact-checking in identifying and correcting these inaccuracies, preventing their spread, and mitigating their impact has become increasingly prominent. As a result, fact-checking has become a key research topic in recent years.

With the emergence of large language models, fact-checking capabilities have been substantially improved across various scenarios. While single-model approaches demonstrate effectiveness in processing straightforward factual queries[12], their limitations become apparent when handling complex information verification tasks. These limitations primarily manifest as model biases and hallucination phenomena, which can compromise the reliability of assessments in nuanced contexts[12]. On the other hand, existing multi-model frameworks and agent-based systems, while attempting to address these challenges, often adopt suboptimal serial processing architectures[9,11]. This structural constraint limits their ability to fully leverage the synergistic potential of inter-agent collaboration and knowledge integration, suggesting significant opportunities for innovation in multi-agent collaborative verification paradigms.

To address the limitations of existing fact-checking models, this study proposes a collaborative fact-checking framework leveraging multi-agent debate and weighted voting. In this framework, multiple heterogeneous agents exchange evidence and perspectives through debate, effectively utilizing their respective expertise and information resources. Furthermore, we introduce a dynamically adjusted weighted voting mechanism. This mechanism autonomously adjusts the agents' weights in the final result based on their performance and credibility, thereby effectively integrating the advantages of heterogeneous agents and enhancing the accuracy and reliability of verification.

The primary contributions of this research are threefold: First, we present a multi-agent fact-checking framework based on group debate, which simulates real-world argumentation processes, improving the comprehensiveness and objectivity of information verification. Second, we combine a voting mechanism with dynamic weight adjustment, proposing a method that effectively integrates the strengths of heterogeneous agents. Finally, we validate the effectiveness of our multi-agent framework using publicly available datasets, demonstrating significant performance improvements in fact-checking tasks, thus proving its efficacy and superiority.

II. Literature Review

A. Fact-Checking

Fact-checking is the process of finding relevant evidence for a given claim and then making a judgment on its veracity based on that evidence[1,2]. While simple claims can be readily verified with singular evidence, as demonstrated by prior models [3,4], the verification of complex real-world assertions necessitates sophisticated methodologies, such as deep learning[5,6]. However, these methods often exhibit limited generalizability due to their inherent reliance on specific datasets and trained models. Although several works utilize large language models to enhance fact-checking performance, these methods are still constrained by the inherent limitations of large language models and their hallucination problems[7, 13].

B. Multi-agents drivend Fact-Checking

The development of LLMs has led to the creation of LLM agents with diverse applications across various domains. For instance, Liang et al. [15] explored a multi-agent debate framework, demonstrating the collaborative problem-solving capabilities of LLM agents. Leveraging these advantages, Fact-Agent is designed to interpret diverse clues and real-world contexts for fake news detection. FACT-AUDIT [11] leverages importance sampling principles and multi-agent collaboration, generates adaptive and scalable datasets, performs iterative model-centric evaluations, and updates assessments based on model-specific responses. Reference [10] proposes the Multi-Agent Debate Refinement (MADR) framework, leveraging multiple LLMs as agents with diverse roles in an iterative refining process aimed at enhancing faithfulness in generated explanations. LoCal [8] primarily consists of a decomposing agent, multiple reasoning agents, and two evaluating agents.

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While most of these existing multi-agent models have yielded promising results, they often suffer from insufficient concurrent knowledge sharing among agents, thus limiting the enhancement of fact-checking effectiveness.

III. Methodology

This section will cover three core aspects: Agent Role Design, System Flow Design, and Prompt Design. The Agent Role Design section will provide a detailed description of the functionalities and interaction patterns of each agent within the system. The System Flow section will elaborate on how the system accomplishes the fact-checking task through multi-agent collaboration. The Prompt Design section will introduce the design of the prompts used to guide agent behavior and decision-making.

A. Agent Role Design

User Agent: The User Agent serves as the interface for user interaction, responsible for receiving user-inputted claims requiring verification. It transmits these claims to the Master Agent and presents the final verification results to the user. Design emphasis is placed on user-friendliness, providing a clear and concise interface to facilitate easy input and result comprehension.

Master Agent: The Master Agent functions as the central control unit, orchestrating and managing the entire verification process. Its primary duties include: receiving user-submitted claims, initiating and managing group discussions among Verification Agents, monitoring the debate to ensure thorough evidence exchange, organizing agent voting post-debate and calculating the final verification outcome based on agent weights, dynamically adjusting agent weights to optimize verification accuracy, and delivering the final verified results to the User Agent.

Verification Agents: The Verification Agents are the operational core, executing the actual fact-checking tasks. A odd number of Verification agents are deployed, to ensure voting result reliability. Each agent possesses varied knowledge bases and reasoning capabilities, enabling multi-perspective claim assessment. Their responsibilities include: participating in debates, exchanging evidence and perspectives, conducting fact-checking based on their knowledge and reasoning, voting according to verification results and credibility, and receiving and adapting to dynamically adjusted weight information from the Master Agent.

B. System Flow Design

As shown in Figure 1, the system framework comprises one User Agent, one Master Agent, and three Verification Agents.

The system's operational workflow is as follows:

- Information Input: The user inputs information requiring verification through the User Agent.
- Debate Initiation: The Master Agent receives the information and initiates a group discussion (group chat) among the Verification Agents.
- Evidence Exchange: The Verification Agents engage in debate, exchanging evidence and perspectives.
- Fact-Checking: The Verification Agents independently perform fact-checking based on their knowledge and reasoning.
- Voting: The Verification Agents cast votes based on their verification results and assigned weights.
- Result Aggregation: The Master Agent aggregates the votes, using the weighted voting mechanism, to calculate the final verification result.
- Result Output: The User Agent displays the final verification result to the user.
- Weight Adjustment: The Master Agent dynamically adjusts the Verification Agents weight based on the agents historical verification result.

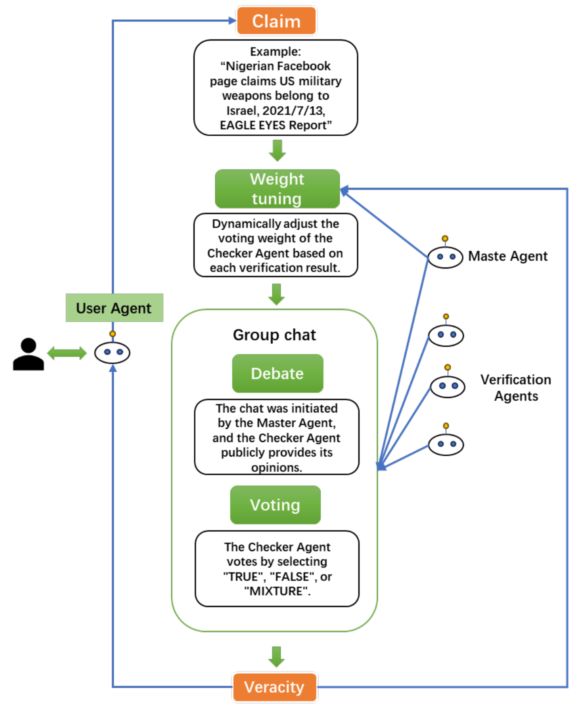

Fig. 1. System framework and operating flow

C. Dynamic Weight Adjustment Mechanism

The dynamic weighting mechanism operates through a dual validation framework: For performance evaluation, the system continuously monitors each agent's verification accuracy (calculated via precision and recall metrics against annotated ground-truth data) and response latency (with predefined maximum processing thresholds). Credibility assessment combines inter-agent consensus scoring (measuring conclusion consistency across multiple agents) with periodic human-in-the-loop validation (random sampling of 10% outputs). Weight adjustments follow an adaptive protocol: agents exceeding upper performance thresholds receive progressively increasing weights, while those below lower thresholds incur exponential decay penalties. The mechanism's robustness is ensured through cross-validation across domains (political/scientific/commercial claims) and complexity levels (simple facts vs. multi-premise inferences), with temporal sensitivity analysis for time-dependent assertions.

D. Prompt Design

To effectively facilitate the multi-agent collaborative fact-checking process, a set of carefully designed prompts are employed to guide the interactions and reasoning of each agent within the system. These prompts are structured to ensure clarity, conciseness, and accuracy in both communication and decision-making:

System message for master agent:

"You are the master control of a fact-checking system, and you are responsible for managing the entire process."

System message for verification agent:

"You are a fact-checking agent, responsible for analyzing information and determining its accuracy."

Prompt for group chat:

"You are now in a group chat. Please determine the accuracy of the given factual information. First, state whether the fact is 'TRUE', 'FALSE', or 'MIXTURE', and provide your reasoning. Keep your statements concise. Here is the fact to be judged:"

Prompt for voting:

"After the discussion, please carefully consider the statements of others and cast your final vote on the given factual information (note: do not judge the statements of others themselves). Choose TRUE, FALSE, or MIXTURE. No other output is needed."

IV. Experiments and Results

In this section, we present the experiments conducted to evaluate the performance of our proposed multi-agent fact-checking framework. We detail the experimental setup, including the datasets used, the evaluation metrics, and the baseline models for comparison. Furthermore, we provide a comprehensive analysis of the experimental results, highlighting the effectiveness of our approach and discussing the implications of our findings.

A. Datasets

The datasets we use is collected from ClaimsKG[14], a structured database which serves as a registry of claims. The latest release of ClaimsKG covers 74066 claims and 72128 claim reviews. The data was scraped in January of 2023 containing claims published between the years 1996-2023(Jan 31) from 13 factchecking websites and ClaimsKG unified these ratings into four categories: TRUE, FALSE, MIXTURE, and OTHER(Our analysis is limited to the first three categories.). We use the Claims Search Engine to get claims from “Politifact” between the years 2013 to 2023. After a filtering phase, our datasets contained 15600 claims rated as TRUE (1,344 statements), FALSE (7,082 statements) or MIXTURE (7,018 statements).

B. LLM model selection

In our experiments, we employed a variety of large language models (LLMs) to drive the different agents, aiming to evaluate the performance of our proposed multi-agent fact-checking framework. The LLMs utilized include:

Qwen 2.5 72B: This is a large language model from the Alibaba Cloud's Qwen series, with 72 billion parameters. Qwen 2.5 demonstrates strong performance in processing Chinese text, exhibiting powerful language understanding and generation capabilities.

GPT-3.5 Turbo: This model, developed by OpenAI, is known for its rapid response times and relatively lower cost. GPT-3.5 Turbo has shown good versatility in handling various natural language processing tasks.

GPT-4o-mini: GPT-4o is the latest model from OpenAI, greatly improved in reasoning, and multi-modal capabilities. GPT-4o-mini is its lightweight version.

These LLMs were accessed via API calls.

In our multi-agent fact-checking framework, we selected appropriate LLMs based on the roles and task requirements of different agents. Specifically:

The Master Agent utilized Qwen 2.5 72B Instruct, leveraging its strong planning and coordination abilities to optimize the overall verification process.

The Verification Agents employed a mix of Qwen 2.5 72B, GPT-3.5 Turbo and GPT-4o-mini, capitalizing on their strengths in language understanding and information retrieval to conduct more comprehensive fact-checking.

Through this approach, we aim to explore the performance of different LLMs in multi-agent collaborative fact-checking and to evaluate the effectiveness of our proposed framework.

C. Main Results

In this section, we present the experimental results obtained from evaluating our proposed multi-agent fact-checking framework. We conducted experiments using established datasets to compare the performance of single large language model (LLM) fact-checking against our multi-agent collaborative approach. Detailed experimental results are presented in Table 1. Notably, the multi-agent implementation was facilitated by the AutoGen framework. Given that the fact-checking task is treated as a three-class classification problem (TRUE, FALSE, MIXTURE), we employed both macro and weighted averages to compute precision, recall, and F1-score.

TABLE I. Exberiment results

Methods	Accuracy	Precision		Recall		F1	
		Micro	*Weighted*	*Micro*	*Weighted*	*Micro*	*Weighted*
Qwen 2.5 72B	0.58	0.51	0.62	0.53	0.58	0.51	0.60
GPT 3.5 Turbo	0.48	0.38	0.49	0.34	0.48	0.32	0.43
GPT-4o-mini	0.54	0.49	0.59	0.51	0.54	0.47	0.55
Our approach	**0.65**	**0.56**	**0.63**	**0.54**	**0.65**	**0.54**	**0.64**

Key Observations:

a)Our Approach Outperforms Individual LLMs: Our approach consistently demonstrates the highest performance across most metrics. This indicates that the multi-agent collaborative framework effectively leverages the strengths of different LLMs, leading to improved overall performance.

b)Qwen 2.5 72B Shows Strong Performance: Among the individual LLMs, Qwen 2.5 72B exhibits the best performance. It achieves relatively high scores in all metrics, suggesting its strong capabilities in handling the fact-checking task.

c)GPT-4o-mini Demonstrates Moderate Performance: GPT-4o-mini shows moderate performance, outperforming GPT 3.5 turbo, but underperforming Qwen 2.5 72B.

d)GPT 3.5 Turbo Exhibits Lower Performance 3.5 Turbo consistently shows the lowest performance across all metrics. This suggests that, in this specific context, its capabilities are less suited for the fact-checking task compared to the other models.

Our proposed multi-agent approach demonstrates a significant improvement over individual LLMs in fact-checking tasks. This suggests that leveraging the diverse capabilities of multiple agents and LLMs through a collaborative framework can lead to more accurate and reliable fact-checking results.

V. Conclusion

This study proposes a multi-agent fact-checking framework leveraging debate and weighted voting, where experiments demonstrate its significant superiority over individual LLMs in verification accuracy and reliability. The framework overcomes single-model biases and limitations of sequential multi-agent systems through dynamic knowledge sharing and heterogeneous agent collaboration, with a dynamically adjusted weighted voting mechanism enhancing decision robustness. While focused on political misinformation to address context-dependent verification challenges, future extensions could adapt the framework to scientific/commercial claims via domain-specific knowledge bases and credibility metrics, while exploring scalability with larger agent networks and cross-cultural debate strategies.

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